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Leonie Hagitte[1,2,3] and Andreas M. Brandmaier[2,3,4]

[1]Goethe University Frankfurt

[2]Max Planck Institute for Human Development, Berlin, Germany

[3]MSB Medical School Berlin, Berlin, Germany

[4]Max Planck UCL Centre for Computational Psychiatry and Ageing Research, Berlin, Germany

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Psychological research frequently is concerned with latent constructs that cannot be observed directly, such as intelligence or personality. These constructs are inferred from manifest indicators, such as item responses or behavioral measures, using measurement models. The quality of these models is central to substantive inference: if a measure does not adequately represent the construct of interest, even highest statistical rigor cannot prevent misleading conclusions (Flake and Fried 2020; Flake et al. 2017). However, recent work has emphasized the need for stronger statistical methods (Flake et al. 2017; Haig 2020; Altoè et al. 2020) but also more explicit theoretical foundations in psychology (Eronen and Bringmann 2021). We argue that measurment quality is essential to both developments. A key aspect of measurement quality is measurement invariance (MI), that is, the assumption that a measurement model functions equivalently across groups, samples, or time points. MI is required when researchers compare latent means, variances, or relations involving latent variables across such conditions, which is the case in typical correlational and experimental research designs. If MI is violated, observed differences may reflect measurement artifacts rather than substantive differences in the construct.

Classical Measurement Invariance Testing

MI testing in the classical sense is usually done using Multi-group confirmatory factor analysis (MGCFA), sometimes also referred to as Multi-group Structural Equation Modeling (MGSEM). Thereby MI testing usually proceeds through a sequence of multiple-group model comparisons in which equality constraints are imposed on increasingly restrictive sets of parameters, such as factor loadings, intercepts, and residual variances. Equality of the models structure in both groups is calles configural Invariance, and can be understood as both groups share the models general form (i.e. number of latent factors and items). Equality of factor loadings former is referred to as metric invariance, if metric invariance is present it indicates that all factor loading parameters are equal across groups. Specifically, this suggests that the items of a given scale and their underlying constructs relate to each other with the same strengths in both groups. Furthermore, it is called scalar invariance if the item intercepts are invariant across groups as well. If scalar MI is established, this suggests that in addition to the factorial structure and the factor loadings, the vectors of the item intercepts are also invariant. Scalar invariance is necessary for the comparison of latent mean differences across groups, because it indicates that the scales that were used have the same unit of measurement as well as the same origin. Residual or strict invariance assumes, additionally to the constraints above, that variables residuals are equal across the groups. If the residual variance holds, the items of the scale are measuring the latent constructs with the same measurement error. This implies that it can indeed be known whether differences measured by the items of the tested scale between the groups are due to group differences in the proposed factors or due to measurement errors. What is only implied in those interpretations of the different MI-levels, is that the question which level specifically one "needs" to hold in ones one data is determined by ones research question and the goal of the inference. As mentioned, metric invariance for example is sufficient to varrant latent mean comparisons between groups. In many empirical studies this is what researchers aim to do, compare latent mean or sum scores of individuals of different groups, then metric invariance suffices as MI-level.

Moderated Nonlinear Factor Analysis

However, MGCFA testing also has some limitations (e.g.: groups need to be created by categorical variables; the differences of said groups are assumed to be linear in their difference) that were overcome with more contemporary approaches to MI-testing. One example, that is getting more popular, is the Moderated nonlinear factor analysis (MNLFA). By conceptualizing MI differently, it enables the possibility to have the groups being created via continuous and potentially nonlinear variables. In MNLFA testing Measurement non-invariance (MNI) is modeled as the moderation of the measurement model (or parts of it) by a variable, external to that measurement model. That moderator variable is the variable, that was the former "grouping variable" in the MGCFA approach. The understanding is that if any model parameters are moderated by that external variable, then the measurement model is not measurement invariant, because it performs differently dependent on that variable.

The exogenous variable may influence any subset of the parameters in the measurement model. Specifically, the means, variances, and covariance of latent factors, as well as the item intercepts, factor loadings, and residual variances of any manifest indicator, may vary systematically across individuals as deterministic functions of the exogenous variable. Whether MI is present depends on which parameters are moderated by the exogenous variable. If moderation is limited to the latent-factor parameters—that is, the means, variances, and covariance — the model remains consistent with MI. In contrast, moderation of item-level parameters, including intercepts, loadings, or residual variances, constitutes MNI. Accordingly, a moderation function must be specified for each parameter that is allowed to vary as a function of the exogenous variable. Because of this conceptualization, the exogenous variable, the moderator, can be categorical as well as continuous, as well as non-linear, or linear.

Structural Equation Modeling Trees